Rigid PCB Manufacturing Capabilities

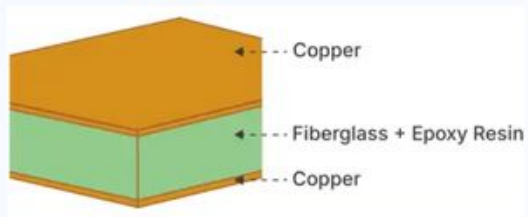
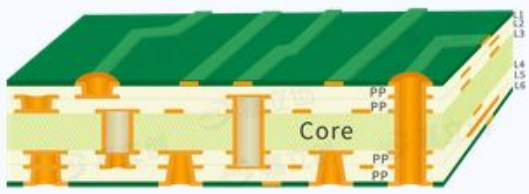
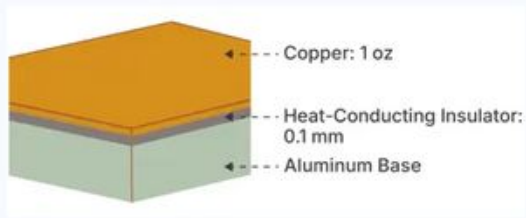
JLCPCB offers 1-32 layer rigid PCB manufacturing and assembly services with various substrate thicknesses, surface finishes, and impedance control.

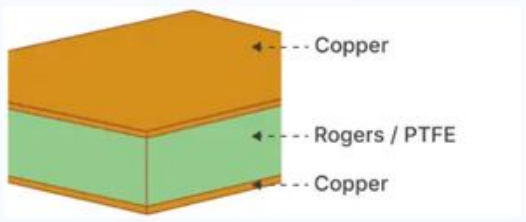
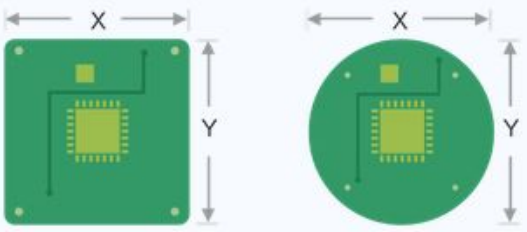
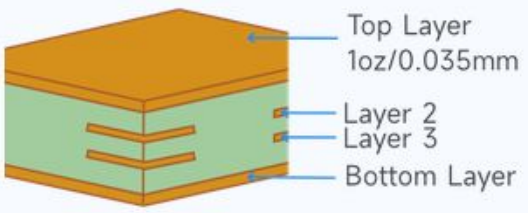
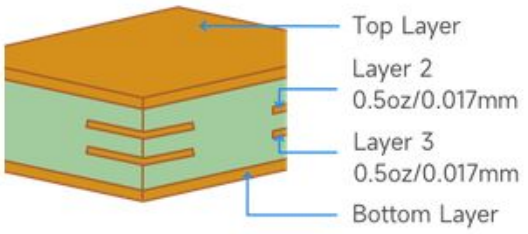
Green Purple Red Yellow Blue White Black

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PCB Specifications

Features	Capability	Description	Patterns
Layer count	1-32 Layers	The number of copper layers in the PCB	
Controlled Impedance	4/6/8/10/12/14/16/18/20/.../32 layers	User Guide to the JLCPCB Impedance Calculator JLCPCB Impedance Calculator	
Impedance Tolerance	±10%		
Material	FR-4	Grade A laminates from suppliers including Nan Ya, KB, Shengyi and etc.	
	HDI	Currently support (1-step / 2-step / 3-step) ① Blind Via Diameter: 0.075mm-0.15 mm (fabricated by laser drilling and copper-filled electroplating process) ② Buried Via Diameter: - Standard range: 0.15 mm – 0.55 mm (fabricated by mechanical drilling with Epoxy Filled & Capped) - Extreme capability: Minimum buried via down to 0.10 mm, applicable only when the dielectric thickness penetrated by mechanical buried via ≤ 1.0 mm. ③ Minimum trace width / space: 3 mil / 3 mil; Extreme capability: 2.7 mil / 2.7 mil. ④ Minimum dimension: 5 × 5 mm; Maximum dimension: 576 × 469 mm. ⑤ HDI base material: Nan Ya NP-175F TG170/Shengyi S1000-2M TG170. For more details, please refer to HDI PCB capabilities FAQ	
	Aluminum-Core	1-layer Aluminum-core PCBs	
	Copper-Core	1-layer copper-core PCBs with direct heatsink contacts to core (≥ 1 × 1 mm)	

		heatsink contacts to core (≥ 1 × 1mm)	
	RF PCB	1 oz copper, 2-layer RF PCBs with Rogers and PTFE cores	
FR-4 Dielectric Constants	4.5 (2-Layer PCB)	7628 Prepreg 4.4 3313 Perpreg 4.1 2116 Perpreg 4.16	
Maximum Dimensions	FR4(1-layer): 606 × 510 mm FR4(2-layer): 670 × 600 mm FR4(4-layer): 663 × 593 mm FR4(6-layer and above): 656 × 586 mm Rogers / PTFE Teflon PCB: 590 × 438 mm Aluminum PCB: 602 × 506 mm Copper PCB: 480 × 286 mm	These limits apply to PCBs with thickness ≥ 0.8 mm. The thinner FR4 PCBs are 599 × 497 mm maximum. 2-layer FR4 PCBs can reach a maximum size of 1020 × 600 mm. 4-layer FR4 PCBs can reach a maximum size of 1016 × 596 mm.	
Minimum Dimensions	FR4/Rogers/PTFE: 3 × 3 mm Castellated / Plated Edges: 10 × 10 mm Aluminum/Copper PCB: 5 × 5 mm	These limits apply to PCBs with thickness ≥ 0.6 mm. Manual review required for thinner PCBs. Panelization is recommended for small-sized boards.	
Dimension Tolerance	±0.1mm	±0.1mm(Precision) and ±0.2mm(Regular) for CNC routing, and ±0.4mm for V-scoring	
Thickness	0.4 – 4.5 mm	Thickness for FR4 are: 0.4/0.6/0.8/1.0/1.2/1.6/2.0 mm (2.5 mm and above are for 12+ layer PCBs only)	
Thickness Tolerance (Thickness≥1.0mm)	± 10%	e.g. For the 1.6mm board thickness, the finished board thickness ranges from 1.44mm(T-1.6×10%) to 1.76mm(T+1.6×10%)	
Thickness Tolerance (Thickness<1.0mm)	± 0.1mm	e.g. For the 0.8mm board thickness, the finished board thickness ranges from 0.7mm(T-0.1) to 0.9mm(T+0.1).	
Finished Outer Layer Copper	2-layer: 1 oz / 2 oz / 2.5 oz / 3.5 oz / 4.5oz Multi-layer: 1 oz / 2 oz		
Finished Inner Layer Copper	0.5 oz / 1 oz / 2 oz	Finished copper weight of inner layer is 0.5oz by default.	
Soldermask	Green, Purple, Red, Yellow, Blue, White, and Black.	We use LPI (Liquid Photo Imageable) solder mask. This is the most common type of mask used today. Heat-cured ink soldermask is usually found on low-cost, single-sided PCBs.	
Surface Finish	HASL (leaded / lead-free), ENIG, OSP (Single-sided FR-4, FPC, and aluminum PCBs are currently not supported for OSP)	Aluminum PCBs: HASL only; FR-4/HDI boards with 6 or more layers, boards with a thickness ≤ 0.4 mm, high-frequency boards, copper-based PCBs, and FPCs do not support HASL.	

Drilling

Traces

Soldermask

Legend

Outline

FAQs about Rigid PCB Capabilities

What is the maximum number of layers supported by JLCPCB for standard production?

JLCPCB supports up to 32 layers in standard production. Layers from 1 to 6 offer the optimal balance between cost, lead time, and reliability. For designs exceeding 10 layers, we recommend submitting the stack-up diagram for engineering review prior to ordering to ensure manufacturability and controlled impedance performance.

Does JLCPCB provide impedance control, and what tolerance can be achieved?

What are the key differences in manufacturing capabilities between standard and advanced options?

Which factors most significantly affect PCB board thickness and how can it be controlled?

How does panelization affect pricing and manufacturing quality?

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